CSL213: **Data Structure and Program Design – I**

Lab Assignment (Announcement: 31st October, 2022)

Due Date : 7th November, 2022 (By 12:00 noon)

For batch - R1

Notes

1. You can choose to do the assignment in a group of 2 students. If done in a group, both students should be from the same practical-batch.
2. No late submission will be allowed.
3. No copying/sharing the code across the groups. If found copied, NO marks will be awarded to the assignment for all such groups including original authors.
4. Evaluation will be done during 7th November to 11th November, 2022. Please make sure that you attend your lab hours during this week. For some batches, the evaluation may take place at scheduled time other than lab hours as will be announced.
5. You need to submit your source files by
    - Attaching these files in a mail by sending the mail to *dspd.assignment@gmail.com*

   Please attach only *.c, *.h files (and other i/p text files if any). *NO *.objs and *.exe to be attached.*

6. Please mark your subject line as *DSPD-1-Assignment: Your Enrollment numbers.*

Cricket Tournament:

There were N (N is even) teams participated in a cricket tournament. All teams were divided into two groups having equal number of teams. Each team in a group played in Round Robin fashion (each team played with other teams in the same group). Winning team will get 2 points and no result/tie gives 1 point to each team. Two top teams from each group lead to semi-finals, which always be the case to be assumed. Finals will be between the winners of the two semi-finals.

Define three structures named as ***player***, ***team*** and ***match_played***.

A. ***player*** attributes- player_id, match_id, previous_total_score, previous_avg, previous_total_wickets, previous_wicket_avg, total_runs_in_last_five_matches, total_wickets_in_last_five_matches, century, present_match_score, present_match_wicket, player_role, out/not out. The attribute player_role identify the player as batsman, bowler or all-rounder.
B. ***team*** (set of players) - Each team has 15 players.
C. ***match_played*** attributed - match_id, teams_played, higest_run, man_of_the_match, wicket_taken_by_pacer, match_result (winning team/no result/tie) .

After all the matches played in the tournament, find out-

i. The possible combination of playing 11 to form a team following rules are to be maintained:
    - Minimum number of batsman= 5
    - Minimum number of bowler= 4
    - Minimum two best batsmen (on the basis of maximum runs and highest avg)
      [**hint**: sort the players based on the key < previous_total_score , previous_avg >]
    - Minimum two best bowlers (on the basis of maximum wickets and highest avg). If there are multiple such combinations, list them all.

ii. Highest total individual run getter before knock out matches. If there are multiple such players, sort them based on the runs scored. If there are multiple players scoring same number of runs, then sort on the basis of their names.

(P.T.O.)

iii. The player who was declared as man_of_the_match for maximum number of times. If there are multiple such players, sort the output based on their names.

iv. Check whether results of ii) and iii) are same or not.

v. Calculate highest average by an individual

vi. Difference in total wickets taken between all pacers and spinners of the teams

vii. Find out the players who are declared as man_of _match at-least k times (k is input) and also having maximum number of centuries among such players. If there are multiple such players, sort the output based on their names.